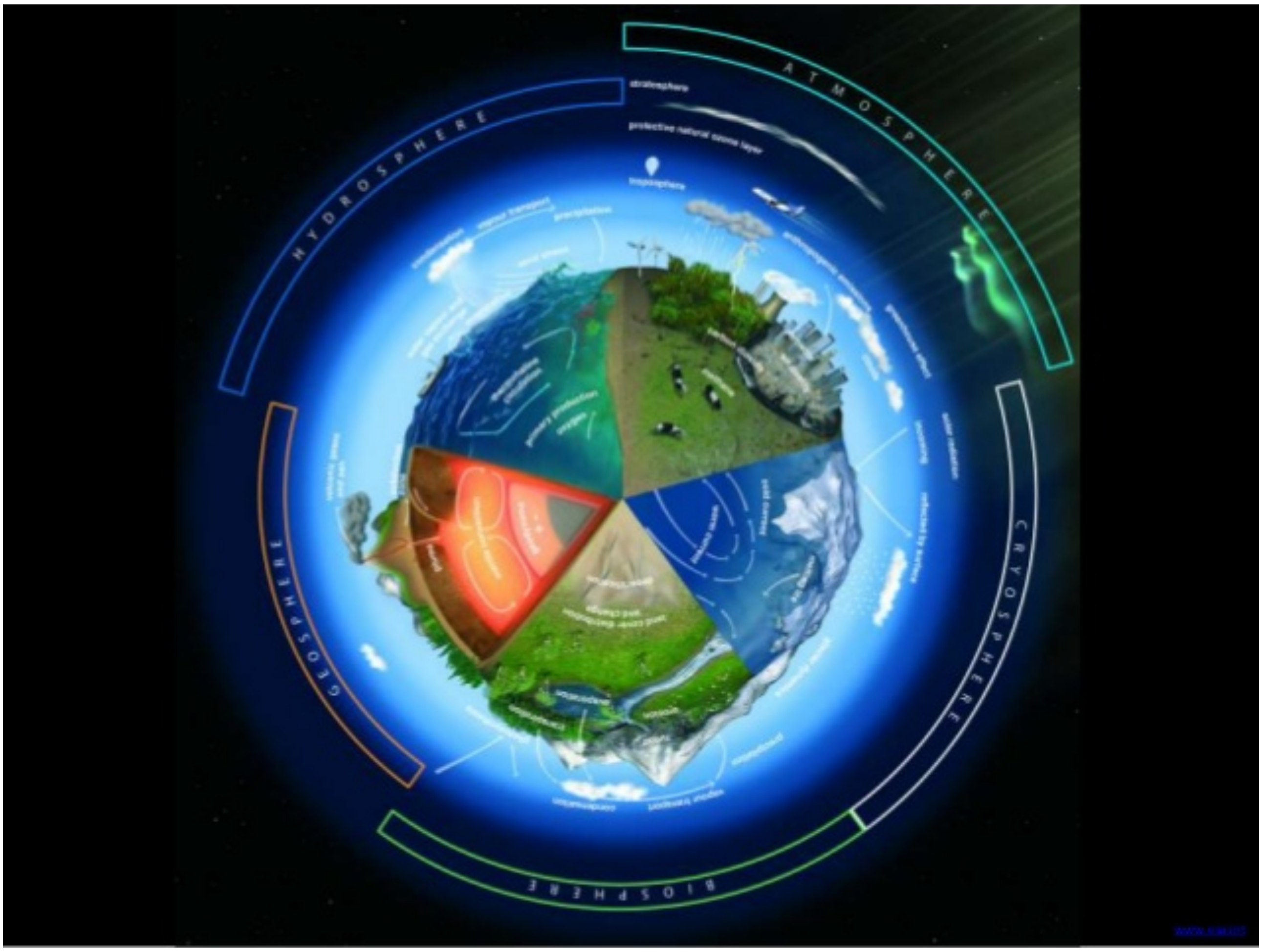
YEAR 10 SCIENCE

EARTH AND SPACE SCIENCE
(Adjusted)

GLOBAL SYSTEMS

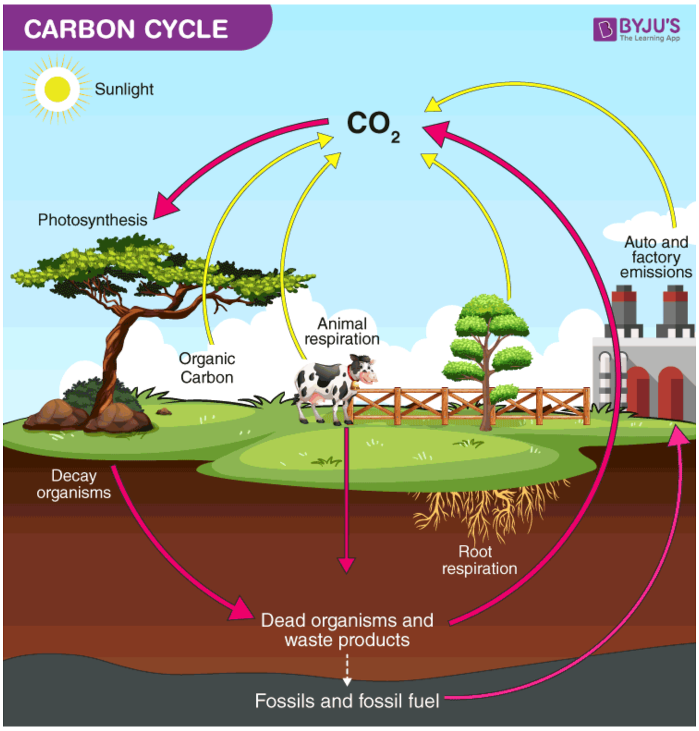
NAME: ______________________________



CYCLES IN NATURE

Elements such as carbon, oxygen, hydrogen, nitrogen and sulphur are recycled in nature - this is the role of the decomposers.

Carbon Cycle



The element carbon is one of the main materials in sugars, proteins and fats - these are found in the tissues of all living things. Carbon is also found in vitamins, as well as in waste products such as CO_2 and urea.

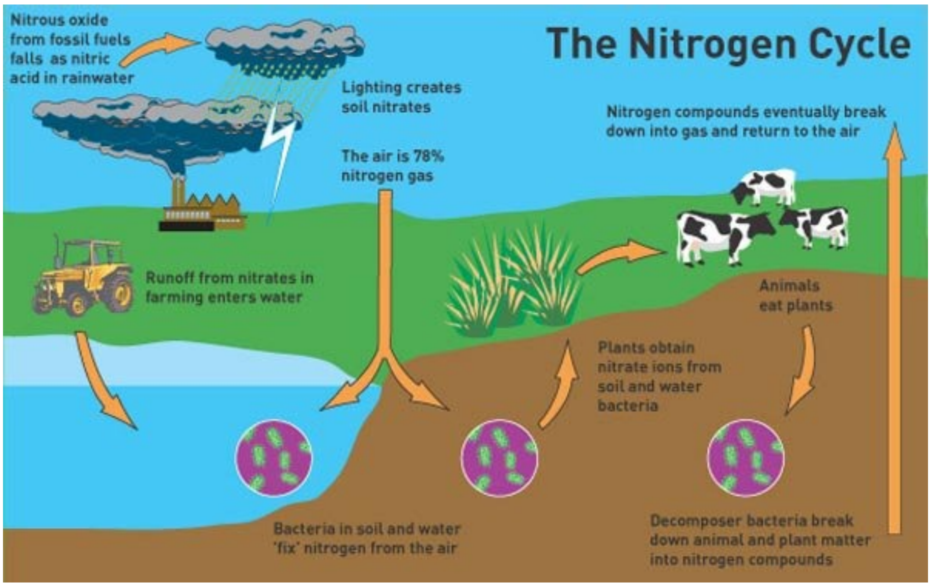
In the carbon cycle, plants convert CO_2 to food and O_2 during photosynthesis.

Animals (and plants at night) produce CO_2 during respiration.

On decomposition, all plant and animal material is reduced to CO_2 by the decomposers, and the residual nutrients are returned to the soil.

Nitrogen Cycle

Nitrogen makes up 78% of the atmosphere, is found combined with oxygen as nitrates in the soil, and is one of the elements found in proteins (these are the building blocks of cells and tissues).



Animals obtain their nitrogen either from plants or other animals.

Plants get their nitrogen from the nitrates in the soil. They are soluble in water and are easily absorbed by plant roots.

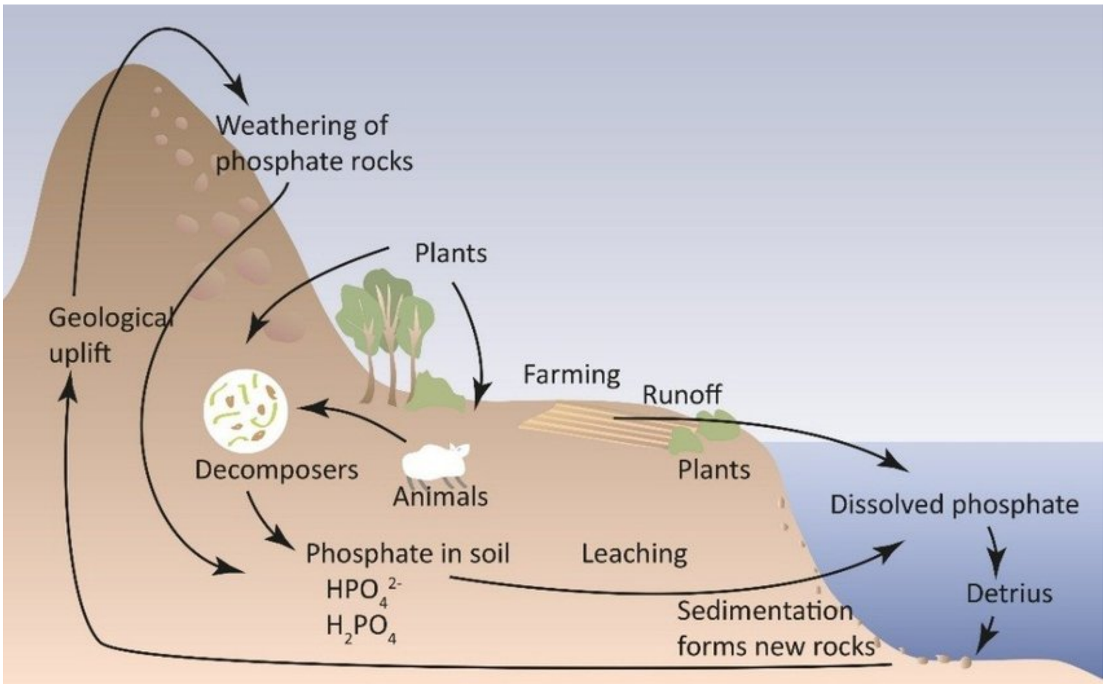
Nitrates can be formed in the following ways.

- The ***decay of dead organisms*** by decomposers.
- Atmospheric nitrogen changed by ***nitrogen-fixing bacteria*** in the soil and in ***nodules*** on the roots of ***legumes*** such as clover, peas and beans.

- ***Lightning*** causes atmospheric nitrogen N_2 to react with O_2 to form oxides of nitrogen. These form nitrates when they dissolve into water.

Phosphorus Cycle

The phosphorus cycle models how phosphorus moves from the lithosphere to the hydrosphere and then through food chains and back.



Phosphorus is essential for plant and animal growth, as well as the health of microbes inhabiting the soil, but is gradually depleted from the soil over time. The main biological function of phosphorus is that it is required for the formation of nucleotides, which comprise DNA and RNA molecules. Specifically, the DNA double helix is linked by a phosphate ester bond.

Calcium phosphate is also the primary component of mammalian bones and teeth, insect exoskeletons, phospholipid membranes of cells, and is used in a variety of other biological functions.

The phosphorus cycle is an extremely slow process, as various weather conditions (e.g. rain and erosion) help to wash the phosphorus found in rocks into the soil. In the soil, the organic matter (e.g. plants and fungi) absorb the phosphorus to be used for various biological processes.

Nature’s time machine

Student: .... Class: ....

Secrets in the ice

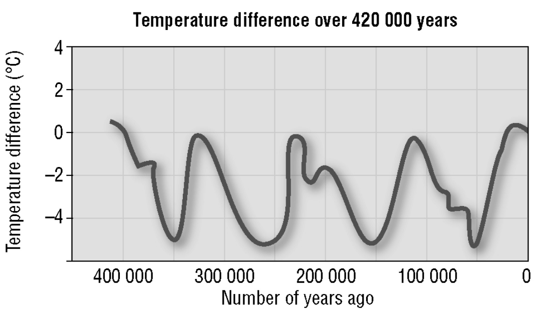
Scientists may not have discovered a time machine yet, but they do have a way of looking back over 300 000 years by looking at **ice cores**. Scientists drill into the ice of Antarctica and collect samples from deep down. Ice cores provide information about the Earth and the atmosphere over hundreds of thousands of years.

For thousands of years, snow has fallen in Antarctica. The snow turns to ice, which builds up over time. Dust, gases and other substances from the air become trapped in the ice. The trapped substances provide information about what was in the air at the time the snow fell.

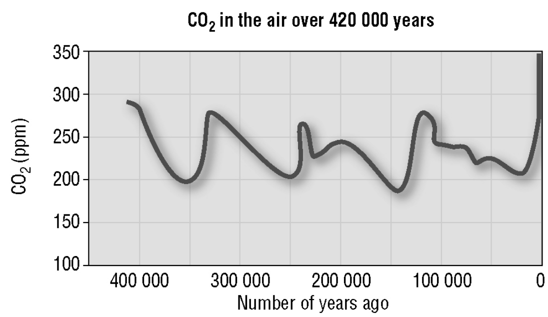
By studying dust in the ice cores, scientists can work out when certain volcanoes have erupted. They have linked radioactive substances in the ice to nuclear disasters in various parts of the world. The dust and radioactive substances were probably carried to Antarctica by the wind.

Scientists have used ice cores to track the amount of carbon dioxide in the air and the Earth’s temperature. The graphs on this page show how these have changed over the past 400 000 years.

This ice core was drilled from more than 3.7 km down. Parts of it are more than 150 000 years old. Photo: Public domain



Zero on the temperature scale represents the Earth’s average temperature now. On the temperature line, the peaks represent warmer periods of time. In between the peaks, the average air temperature was up to 6°C lower than now. These colder times are called the ice ages.



The line for carbon dioxide resembles the temperature line. When the temperature was higher, there was more carbon dioxide in the air. When the temperature dropped, so did the amount of carbon dioxide. (ppm means parts per million.)

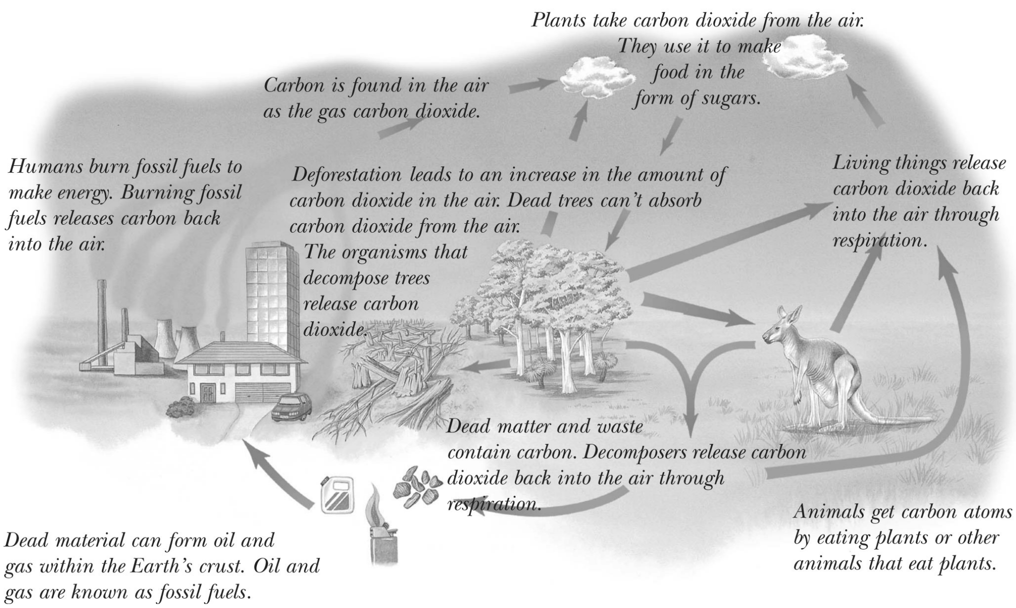
Look at the graphs above of how carbon dioxide and temperature have changed over time and answer the following questions.

1. How many ice ages have there been in the past 400 000 years? ....
2. At a point 350 000 years ago:
   (a) how much carbon dioxide was in the air ....
   (b) how much colder was the Earth’s air compared to the present temperature?..............................
3. Has the Earth ever been warmer than it is now? If so, how much warmer?
   ....
4. When was the amount of carbon dioxide in the air the highest? ....

5. Why is an ice core like a time machine? ....
....

Recycling carbon

Carbon atoms are found in the air, soil, rocks, oil and gas below the Earth's surface, and all living things. The number of carbon atoms on Earth always remains the same. When living things use carbon atoms, the atoms don't just disappear; they are recycled. The diagram below shows the recycling of carbon. It also shows how humans have added to the amount of carbon dioxide in the air. We must take care not to interfere with the natural recycling process.



1. How do animals get carbon as part of the carbon cycle?
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2. Explain two ways in which carbon that is contained within a plant can end up back in the air.
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3. Where in the carbon cycle is carbon stored for long periods of time?
....

4. Would the carbon cycle still take place if humans did not exist? Explain.
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5. How do humans add extra carbon dioxide to the air?
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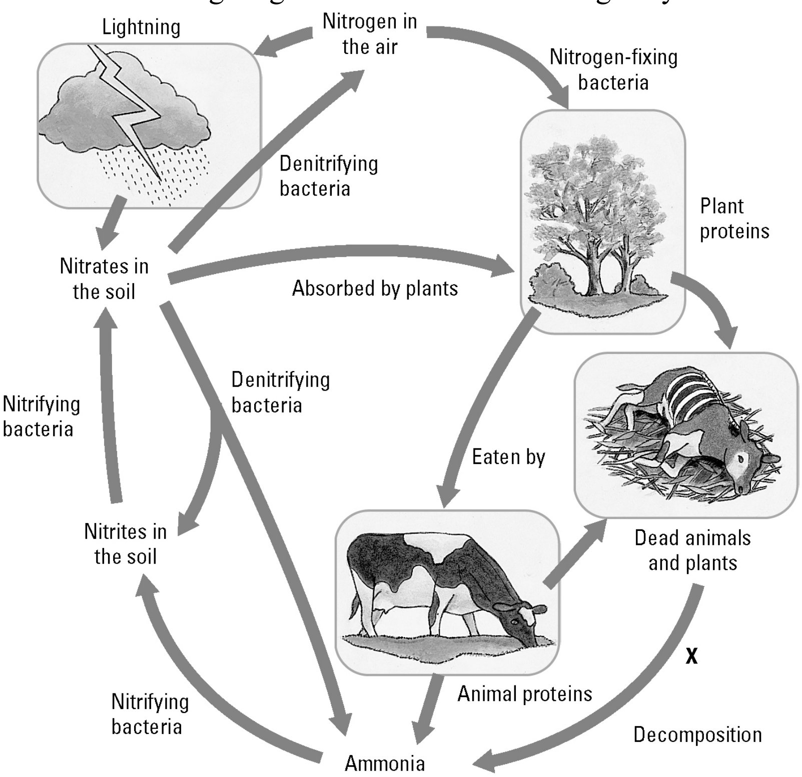
Cycles in nature

Student: .... Class: ....

The nitrogen cycle

1. The following diagram summarises the nitrogen cycle in nature.



The nitrogen cycle

(a) Explain why nitrogen is such an important chemical in nature.

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(b) Identify two decomposers at point X.

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(c) Identify the ways that ammonia can enter the soil.

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(d) Explain why ammonia does not build up in the soil.

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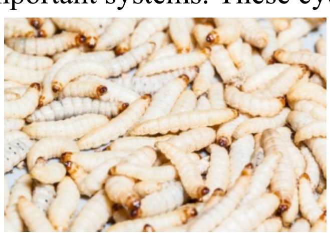
The recyclers

2. Matter and energy are recycled in nature. The nitrogen cycle, carbon cycle and water cycle are very important systems. These cycles involve living things as well as non-living things.



(a) Maggots

(b) Bacteria

(c) Fungi

Photos:
Maggots: Dario Lo Presti / 123RF
Bacteria: dtkutoo / 123RF
Fungi: jarous / 123RF

Explain the role of each of the decomposers shown in terms of recycling matter.

....

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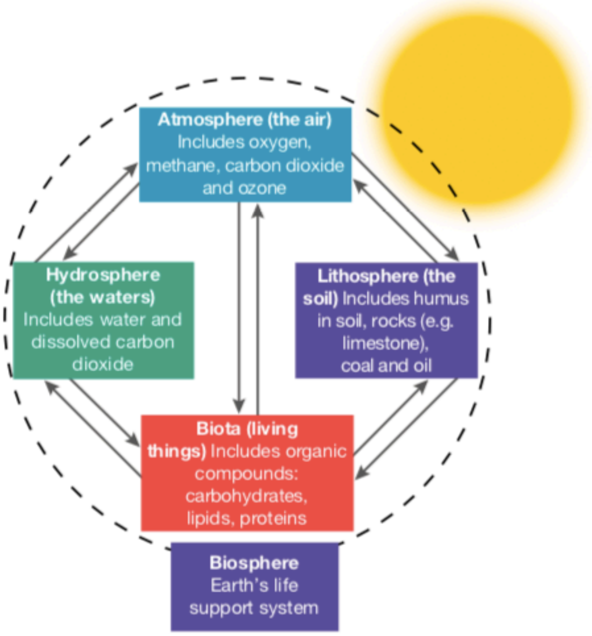
Biosphere

The biosphere consists of the ***atmosphere***, ***lithosphere***, ***hydrosphere*** and ***biota*** (living things), the interactions between them, and the radiant energy of the sun.

The biosphere includes all of the ecosystems on Earth. Interactions within the biosphere include the ***cyclical movement of essential elements*** such as carbon, nitrogen and phosphorus.

The biosphere can be considered Earth's life-support system.



Atmosphere

The Earth's atmosphere is divided into the ***troposphere*** (lower atmosphere) and the ***stratosphere*** (upper atmosphere).

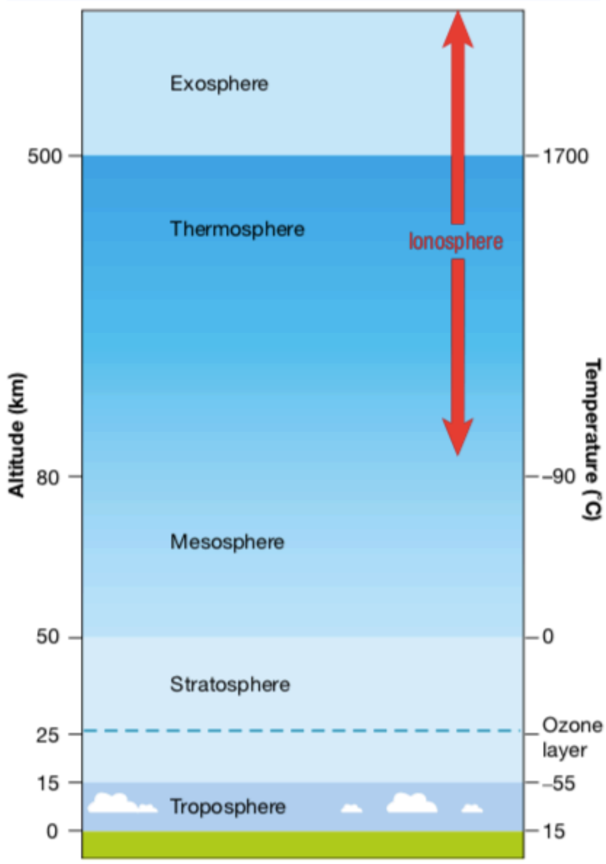
The troposphere is around 6-17 km high depending on your latitude (how close you are to Earth's equator or the poles).

The stratosphere is about 50 km thick and contains an area known as the ***ozone layer***. While this layer allows visible and infra-red radiation from the sun through, it absorbs ultraviolet (UV) radiation. This reduces the amount of damaging UV radiation reaching Earth's surface.

Chlorofluorocarbons (CFCs) have been used as cooling agents in refrigerators and air conditioners, as propellants in aerosols, and as industrial solvents.

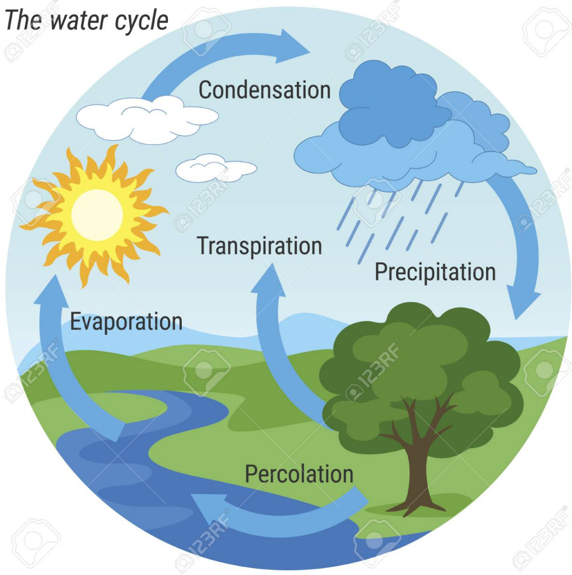
Their use has increased the amount of these compounds being released into the atmosphere. Once in the stratosphere, they are broken down into chlorine atoms, ***which destroy ozone molecules***. This has

Layers in the Earth's atmosphere



depleted areas of the ozone layer, ***increasing the amount of damaging UV rays that get through*** and cause damage to living organisms.

Hydrosphere



The waters of our planet make up the hydrosphere. The simplified figure of the ***water cycle*** shown at right describes how water moves through the biosphere.

Toxic or industrial wastes and untreated sewage have made their way into rivers, bays and the ocean, which has had a direct impact on the hydrosphere. Toxins can move along food chains, in some cases being biologically magnified - getting more concentrated - as they move up the chain.

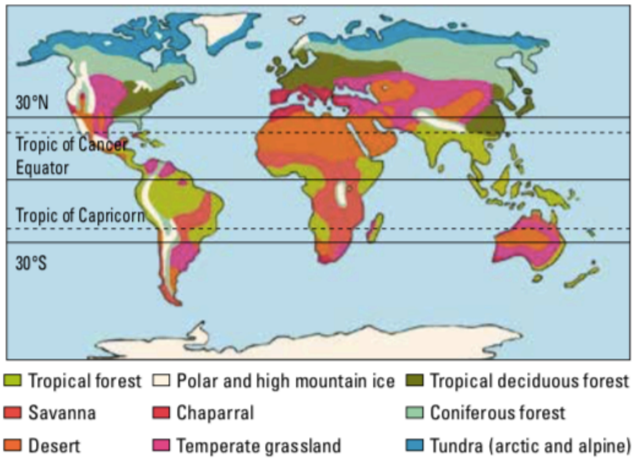
While some of these wastes are ***purposefully dumped***, in other cases they enter the water system in ***run-off from the land*** or are ***washed out of the atmosphere in rain***.

Geosphere (Lithosphere)

Earth's rocky crust and soil make up the ***lithosphere***. It is within this sphere that ***igneous***, ***sedimentary*** and ***metamorphic rocks*** are formed, broken down and changed from one type to another.

The land surface of our planet is divided into regions called **biomes**. The criterion used to divide regions into biomes is the dominant vegetation type.

The type of dominant vegetation within biomes is influenced by environmental factors.



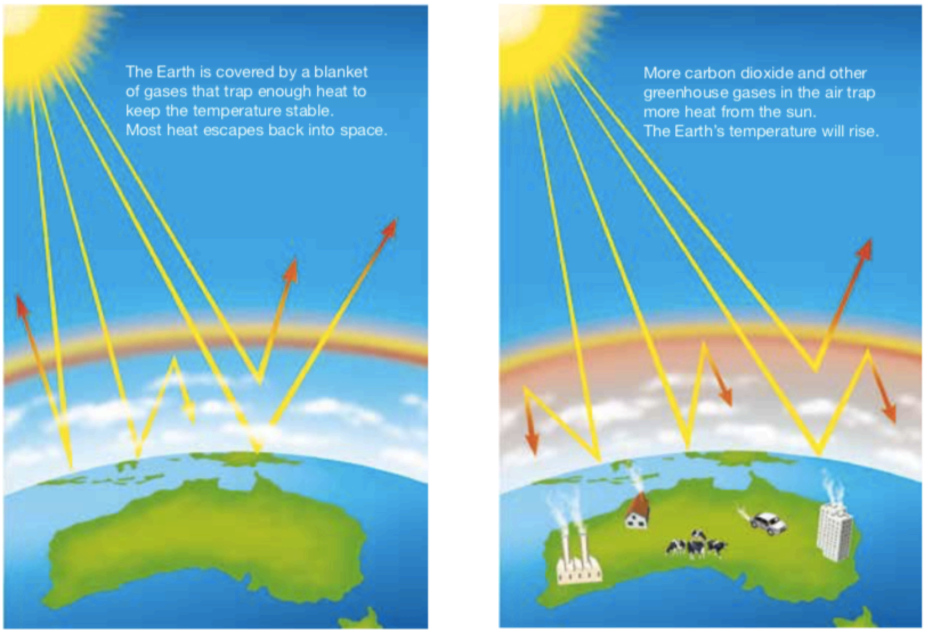
Environmental factors (such as latitude, temperature and rainfall) influence the type of vegetation that can survive in a particular area and so can be used to predict the type of biome that may exist there.

Overstocking, soil exhaustion, salinity, pesticides, unstable landfill, salinisation, toxic seepage, excessive clearing, chemical emissions, deforestation and soil erosion can all be very destructive to the lithosphere. Overgrazing and deforestation may also result in desertification. They can have ***detrimental effects on habitats and resources*** and hence the survival of organisms within the ecosystem.

GREENHOUSE EFFECT

Earth's atmosphere acts like a giant invisible blanket that keeps temperatures on our planet's surface within a range that supports life. Within the atmosphere, greenhouse gases trap some of the energy leaving the Earth's surface to help maintain these warm temperatures. The maintenance of Earth's temperatures by these atmospheric gases is called the ***greenhouse effect***.

Greenhouse gases and the enhanced greenhouse effect

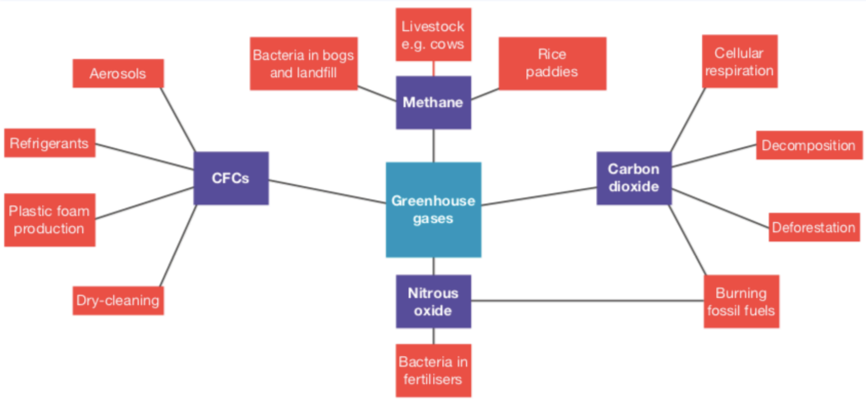


Global Warming

Global temperatures have been increasing and are expected to continue to increase at an accelerated rate. The rising temperature of Earth is known as ***global warming***. This may result in melting icecaps, rising sea levels, increased coastal flooding, unusual weather patterns and ocean currents, and threats to the survival of some living things.

The following diagram summarises the sources of greenhouse gases.

Some sources of greenhouse gases



Sources of Greenhouse Gases

- ### Fossil Fuels

  Scientists assert that ***our increased and growing dependence on fossil fuels*** since the Industrial Revolution of the nineteenth century is a major cause of global warming. They argue that burning fossil fuels (such as coal and oil) has resulted in increased levels of ***greenhouse gases*** (such as nitrous oxide and carbon dioxide) in our atmosphere that are trapping heat, causing the atmosphere to heat up.

  This is referred to as the ***enhanced greenhouse effect***.

- ### Grazing animals

  Animals such as ***cattle and sheep*** produce large amounts of ***methane*** as a waste product.

- ### Bacteria

  Methane is another powerful greenhouse gas and is also produced by the ***action of bacteria*** that live in ***landfills*** and soils used for ***crop production***.

  Much of the nitrous oxide in the atmosphere is produced by the ***action of bacteria*** on ***fertilised soil*** and the ***urine of grazing animals***.

- ### Photosynthesis and Cellular Respiration

  CO_2 is taken from the atmosphere during photosynthesis and released back during cellular respiration.

  This suggests that if producers (plants) are reduced in number or removed from the atmosphere, there will be less carbon dioxide removed from the atmosphere, resulting in an overall increase in this gas.

- ### Decomposition and Fossil Fuels

  Carbon dioxide is also released from dead and non-living parts of ecosystems.

  When these fossil fuels are burned, carbon dioxide is released back into the atmosphere.

- ### Ozone

  Ozone (O_3) in the lower atmosphere is also a significant contributor to the enhanced greenhouse effect.

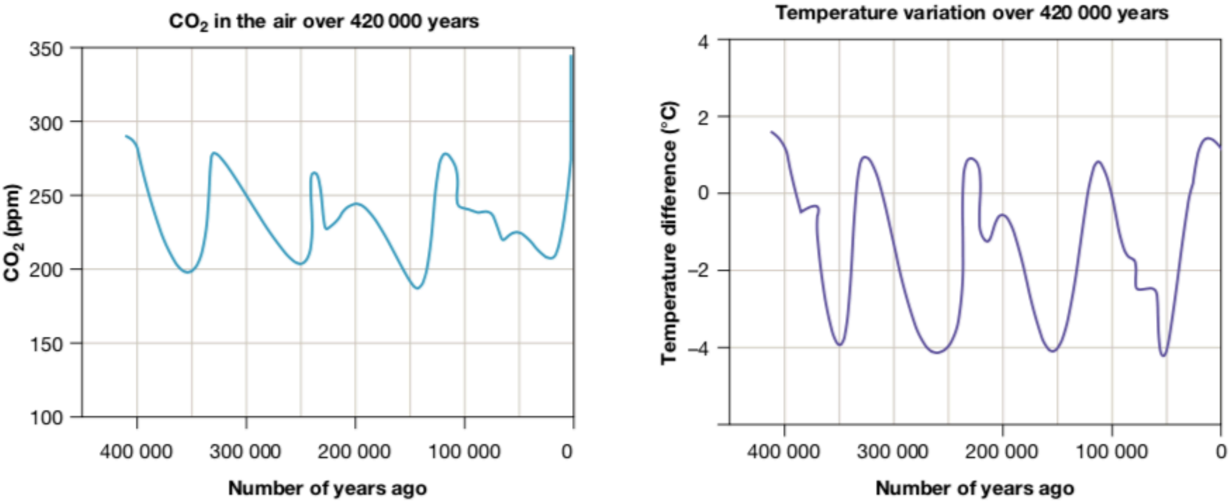
  Although ozone occurs naturally, it is also produced by a photochemical reaction that takes place when sunlight falls on ***emissions from motor vehicles, power stations and bushfires***.

Evidence of Global Warming

- **Increasing CO_2 Levels**

Scientists have used ***ice cores*** to track the air temperature and concentration of carbon dioxide near the Earth's surface in the past. The graphs below show how these have changed over the 420 000 years leading up to the year 2000.

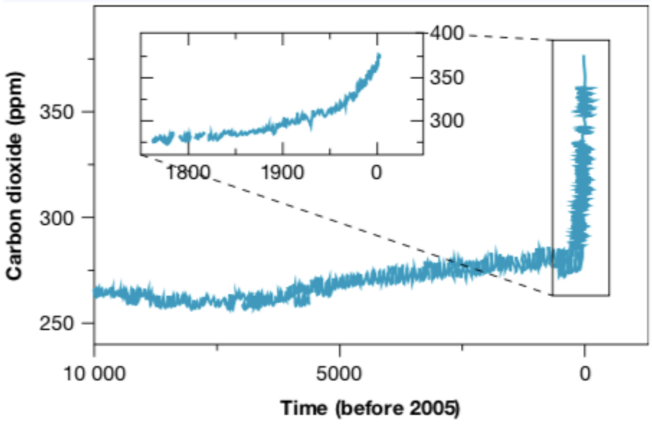
The carbon dioxide concentration is shown in parts per million (ppm) by volume. The temperature difference shown is the deviation from the average temperature now (represented by 0 on the vertical scale). The pattern of changing temperatures resembles the pattern of the change in carbon dioxide concentrations.



During the current decade the concentration of CO_2 has risen to approximately 400 parts per million. There appears to be no significant change in global temperature cycles.

However, the graph below shows that since the Industrial Revolution, there has been a dramatic change in the trend of carbon dioxide in the atmosphere.

This graph shows the dramatic increase in atmospheric carbon dioxide since the Industrial Revolution.



- Climate Models

Meteorologists and other scientists use computer modelling to make predictions about climate change and the possible consequences.

The computer programmes used to model climate change simulate the circulation of air in the atmosphere and water in the oceans. An immense amount of data collected from the atmosphere, ocean and land surface is used, together with mathematical equations that describe the circulation.

- Global Temperature

It is generally agreed that during the next 100 years, the average global temperature could increase by between 1 °C and 4 °C. Computer modelling suggests that the global temperature will not increase evenly across the continents.

According to CSIRO in Australia, temperatures could increase by up to 2 °C by 2030 and up to 6 °C by 2070. As a consequence, there will be:

- more hot days and fewer cold days
- an increase in rainfall in the north-east and a decrease in the south
- more bushfires
- more destructive tropical cyclones

- Rising Sea Levels

According to tide-gauge records, the average global sea level has increased by between 10 cm and 20 cm during the past 100 years.

Sea levels are expected to rise further due to:

- the warming ocean water and its resulting thermal expansion
- the melting of glaciers, the polar ice- caps and the ice sheets of Greenland and Antarctica.

According to NASA, sea ice in the Arctic is melting at the rate of 9 % every ten years. Of the world's 88 glaciers, 84 are receding due to melting ice.

Rising sea levels are likely to cause the flooding of low-lying islands and coastal regions.

- Frozen Soil

Much of the soil on or below the surface of very high mountains in the polar regions is permanently frozen - called ***permafrost***.

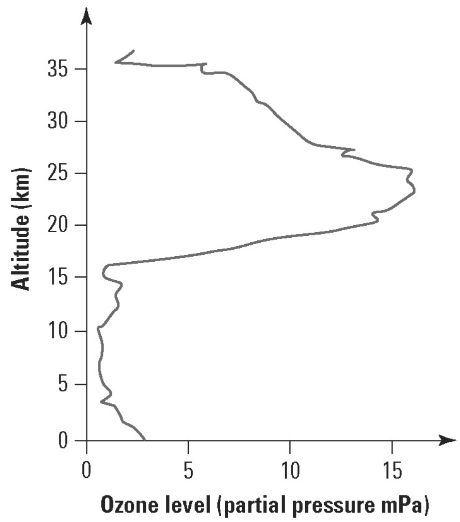
This soil is likely to gradually thaw out as global air temperatures increase, potentially releasing large quantities of CO_2 and methane will be released into the atmosphere. This would increase the rate of climate change.

Ozone layer

Student: .... Class: ....

Ozone and altitude

1. The following graph shows the ozone level (measured as the pressure of ozone in millipascals) as a function of altitude. Answer the questions about this graph.



(a) What is the ozone level at sea level? ....

(b) At what altitude is the maximum ozone level? ....

(c) In what atmospheric layer is the maximum ozone level located?

....

(d) Calculate the ratio of the maximum ozone level in the stratosphere to the maximum ozone level in the troposphere.

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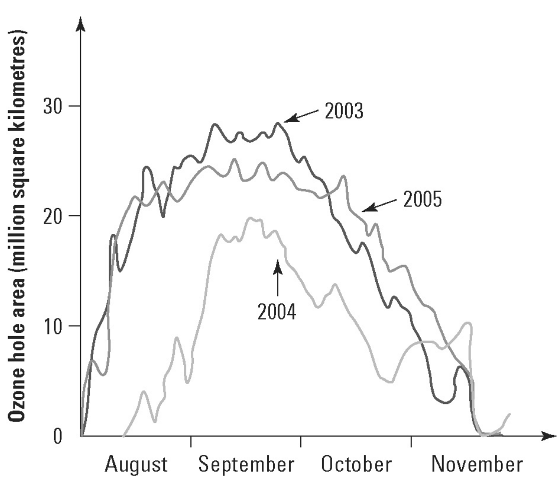
Ozone depletion

2. The following graph shows changes in the area of the ozone hole in the stratosphere above Halley Bay, Antarctica, for the same four-month period over three different years. The ozone hole is a region of the stratosphere that has severely depleted levels of ozone. Answer the questions about this graph.



(a) In which season of the year do the largest ozone holes appear over Antarctica?

....

(b) Compare the area of the ozone holes over these three years.

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(c) Determine the size of the ozone hole at the end of September in 2004.

....

(d) Studies in Chile showed a significant increase in skin cancers since ozone holes first started to appear in the mid-1980s. Explain why this would happen.

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BIODIVERSITY AND CLIMATE CHANGE

The evolution of life forms on Earth has occurred because some organisms are better suited to a particular environment than others.

For some to be better suited than others, there needs to be variation or diversity.

In a global sense, ***biodiversity*** refers to the total variety of living things on Earth, their genes and the ecosystems in which they live.

Earth was a hostile place 3500 million years ago. Fossils provide evidence of structures called stromatolites. They existed in warm sea water and consisted of cyanobacteria, one of the earliest forms of life.

Species Diversity

All environments change over time.

It is the diversity (or variation) of the genetic information within the "gene pool" that contributes to the number of possible combinations that could be used to produce the next generation.

Increased variety in how this genetic information appears in the offspring (i.e. their traits) means an ***increased chance that some of these offspring will be able to survive*** in the environment in which they are born and will live - even if that environment changes.

If there is little variation in the gene pool, there is less chance of the offspring being able to survive possible changes in their environment, such as climate and the availability of habitat, food, mates or other resources.

The consequences of this limited diversity within the population may lead to the ***extinction*** of the species.

Ecological Diversity

Ecological diversity can be considered in terms of the diversity in ecosystems.

The extinction of a particular species within an ecosystem may affect the survival of other species within that ecosystem.

e.g. The extinct species' disappearance will have consequences for the food supplies of others within its food web.

Unless there are other species that can take its place without having a negative effect on others, the survival of other species may be threatened.

Increased biodiversity within ecosystems can reduce the consequences of losing a species to which the survival of others is linked. Likewise, reduced biodiversity in these ecosystems can lead to the extinction of other species.

AUSTRALIA'S BIODIVERSITY

Biodiversity within Australian ecosystems is influenced by both biotic factors (e.g. plants and animals) and abiotic factors.

- Abiotic factors - those that contribute to climate (e.g. temperature and annual rainfall) can affect the abundance, distribution and types of species within a particular ecosystem.

Organisms have particular tolerance ranges for abiotic factors, outside of which they cannot survive.

If global warming results in the development of climatic conditions that are outside a species' tolerance range, and if they are unable to migrate or adapt to the new conditions, then there is a threat that the species may become extinct.

Species that are most at risk are those that have:

- low genetic variability
- long life cycles and low fertility (birth rates)
- a narrow range of physiological tolerance and geographic range
- specialist resource requirements

Global Warming and Australia's Biodiversity

Changes in Australia's biodiversity that may be due to climate change include:

- changes in species' ranges and migration patterns
- shifts in genetic composition of some species that have short life cycles
- changes in lifestyle and reproduction rates

Example

Many plants and their pollinators have ***coevolved*** - they rely closely on each other.

Studies show that climate change has upset the life cycles of pollinators (such as bees). Other studies show that climate change is causing the flowering times of some plants to be out of synchronisation with their pollinators.

With fewer plants being pollinated, fewer are bearing fruit containing seeds essential to produce the next generation of plants.

Preparing to Adapt to Unavoidable Climate Change

The National Climate Change Adaptation Research Facility (NCCARF) has identified eight priority areas for adaptation research. These are:

- terrestrial biodiversity
- primary industries
- water resources and fresh-water biodiversity

- marine biodiversity and resources
- human health
- cities and infrastructure
- emergency management
- social and economic issues

Mass Extinctions

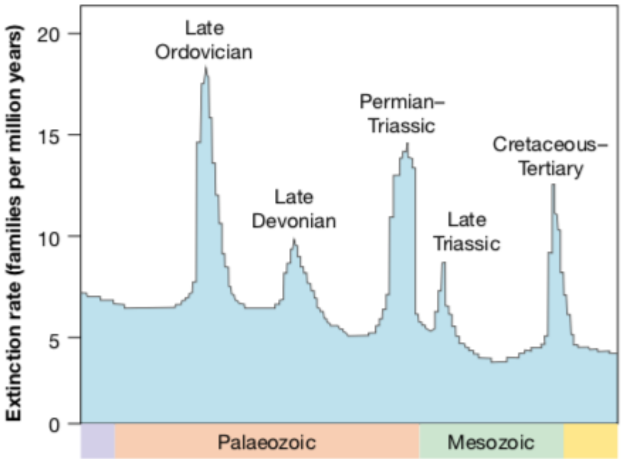
Many scientists believe that we are currently experiencing the ***sixth mass extinction***.

Five other mass extinctions have occurred as a result of global climate change. Some argue that humans are responsible for the current mass extinction.

The International Union for Conservation of Nature has reported that species are ***dying out 1000 to 10,000 times faster than they would without human intervention***.

Those with the view that humans are to blame divide this sixth extinction into two phases.

There have been five mass extinctions in the past — are we currently experiencing a sixth and, if so, is it caused by humans?



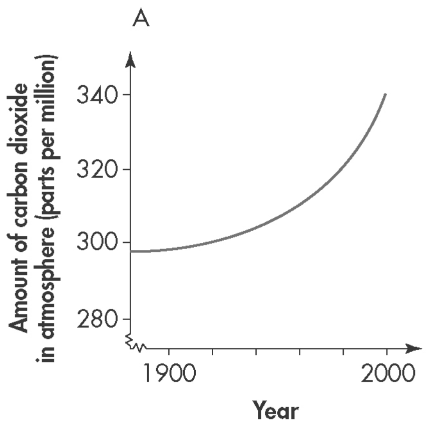
- The first phase began about 100,000 years ago when the first modern humans began to spread throughout the world.
- The second phase began when humans started to use agriculture around 10,000 years ago.

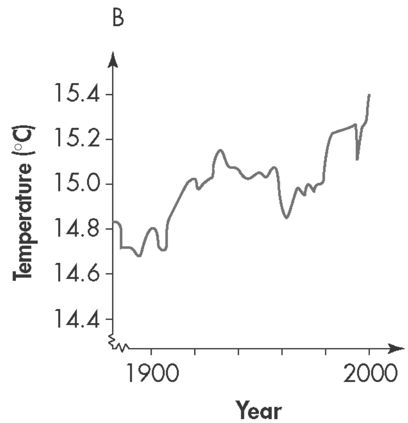
Worksheet 7.4

Global warming

Student: .... Class:..

The two graphs below show information about the atmosphere during the twentieth century.





1. The increase in carbon dioxide levels in the atmosphere over the 100 years of the twentieth century as a percentage is approximately:

A. 14%

B. 40%

C. 297%

D. 337%

2. The percentage increase in atmospheric temperatures over the 100 years of the twentieth century is approximately:

A. 4.7%

B. 14.8%

C. 15.4%

D. 2.35%

3. (a) Identify the major source of carbon dioxide in the atmosphere.

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(b) Explain the connection between carbon dioxide and global warming.

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(c) Outline possible strategies for reducing the amount of carbon dioxide that enters the atmosphere.

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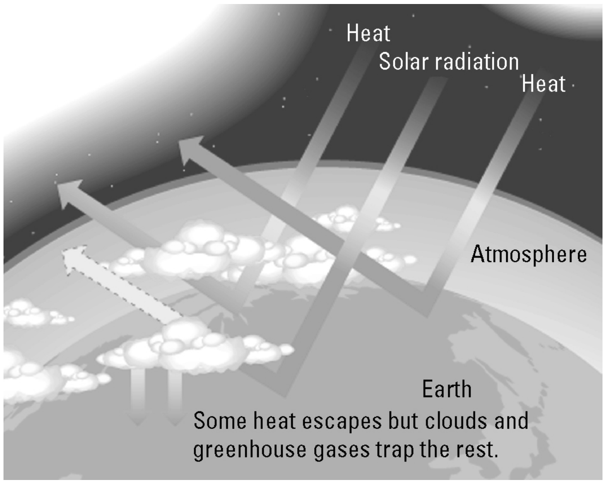
Greenhouse effect



4. (a) Use the graphic on the right to explain how the natural greenhouse effect keeps our planet warm.

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(b) Identify the cause of the enhanced greenhouse effect.

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(c) Explain why the enhanced greenhouse effect is of major concern.

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Carbon dioxide concentrations

5. The following table provides atmospheric carbon dioxide data over recent times.

(a) Plot a line graph of this data and draw the line of best fit.

Year	CO_2 (ppm)
1960	317
1965	321
1970	327
1975	332
1980	340
1985	347
1990	356
1995	362
2000	370
2005	381
2008	386

(b) Global average temperatures have risen during this period. Is this evidence sufficient to state that global warming is caused by increases in carbon dioxide levels? Explain.

....

....

Slowing global warming — alternatives

Student: .... Class: ....

Photo: chuyu / 123RF

Wind power

The wind is free and blows almost continuously; a series of wind turbines can supply a viable and reliable source of energy. It has the added advantage of being non-polluting. However, there has been controversy as wind farms can be quite noisy and some people think they spoil the aesthetics of the natural environment.

Nuclear power

Australia has vast reserves of uranium, the essential element required to produce nuclear power. Enormous amounts of energy can be harnessed through nuclear power generation, but the risks are also great. Waste products and spent nuclear rods remain radioactive for thousands of years, and if a reactor goes into 'meltdown' or these waste materials leak into the environment, the results could be devastating.

Photo: hxdyl / 123RF

Photo: Amornthep Chotchuang / 123RF

Hydro-electric power

Hydro-electricity has been used for decades as an important source of electricity. The natural force of running water turns massive turbines, which produces electricity cheaply. It is also non-polluting. However, river systems must be dammed and this affects the natural environment. Larger tracts of land are often flooded in the process, resulting in the destruction of habitats.

Biosphere 2

More than three decades ago, scientists first raised concerns that the Earth was heading for an 'ecological meltdown'. Their solution was a project known as Biosphere 2. The aim was to replicate the various ecosystems on the Earth, as well as its atmosphere, in a series of domes occupying approximately 1.3 hectares.

In 1991 the project reached fruition, when four men and four women took up residence within the facility for two years. The project was plagued with problems, including the collapse of ecosystems and the demise of several plant and animal species. However, a more serious problem was the inability to sustain an atmosphere; falling oxygen levels and rising carbon dioxide levels meant the project was doomed to fail without external assistance.

Perhaps what scientists really gained was an insight into our future. If we continue to rely on fossil fuels, our climate will continue to change and the atmosphere will continue to deteriorate. We will not simply be able to prop it up from another source. Alternative power sources such as hydroelectricity, wind and nuclear energy are considered sustainable, if somewhat controversial.

Photo: James Mattil / 123RF

1. Explain what is meant by the term 'ecological meltdown'.

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2. What was the aim of Biosphere 2?

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3. What important lessons do you think we can learn from the Biosphere 2 project? Justify your answer.

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4. Complete the following table:

	Advantages	Disadvantages
Hydroelectricity	• Cheap to produce •	• •
Wind power	• •	• Visually polluting •
Nuclear power	• •	• •

5. List some of the questions you would ask an expert in these three alternative energy sources that you feel would help you decide which would be most effective in reducing global warming.

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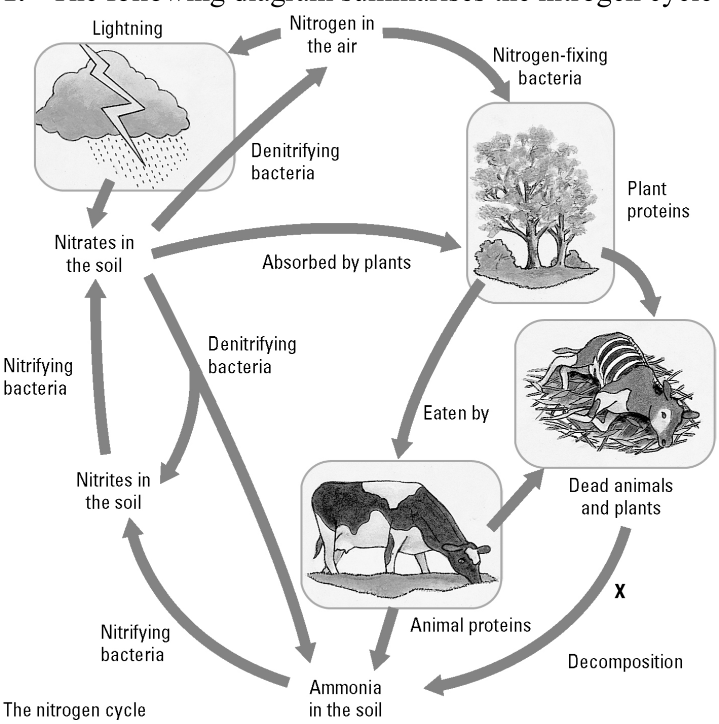
Cycles in nature

Answers

The nitrogen cycle

1. The following diagram summarises the nitrogen cycle in nature.



The nitrogen cycle

(a) Explain why nitrogen is such an important chemical in nature.
Nitrogen is essential for the construction of proteins and DNA and so the nitrogen has to be available in food webs in a suitable nutrient form.

(b) Identify two decomposers at pc
Bacteria, fungus

(c) Identify the ways that ammonia can enter the soil.
Ammonia enters the soil in several ways: due to the action of decomposers; nitrates in the soil being converted by denitrifying bacteria; waste nitrogen from animals

(d) Explain why ammonia does not build up in the soil.
Nitrifying bacteria convert the ammonia into nitrites and then nitrates which are then absorbed by plant roots.

The recyclers

2. Matter and energy are recycled in nature. The nitrogen cycle, carbon cycle and water cycle are very important systems. These cycles involve living things as well as non-living things.


(a) Maggots


(b) Bacteria


(c) Fungi

Photos:
Maggots: Dario Lo Presti / 123RF
Bacteria: dtkutoo / 123RF
Fungi: jarous / 123RF

Explain the role of each of the decomposers shown in terms of recycling matter.

(a) Maggots are the larvae of flies. The flies lay their eggs on dead bodies and so the larvae feed on these bodies and break them down.

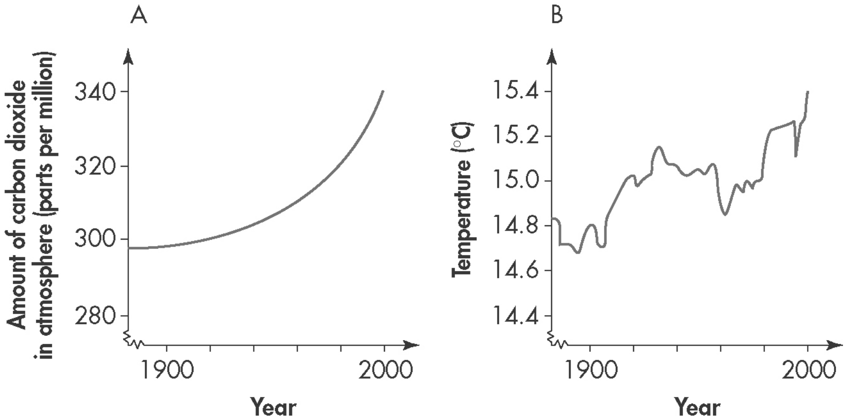
(b) Bacteria will grow on both living and non-living matter. They break down the matter and recycle nutrients.

(c) Fungi feed on dead material. They exude chemicals to decompose dead remains and the nutrients then become available to the ecosystem.

Global warming

Answers

The two graphs below show information about the atmosphere during the twentieth century.



1. The increase in carbon dioxide levels in the atmosphere over the 100 years of the twentieth century as a percentage is approximately:

A. 14%

B. 40%

C. 297%

D. 337%

2. The percentage increase in atmospheric temperatures over the 100 years of the twentieth century is approximately:

A. 4.7%

B. 14.8%

C. 15.4%

D. 2.35%

3. (a) Identify the major source of carbon dioxide in the atmosphere.

Burning of fossil fuels

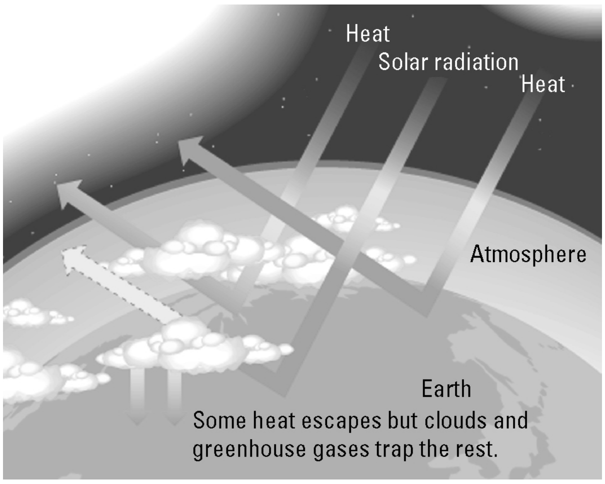
(b) Explain the connection between carbon dioxide and global warming.

Heat radiation from the Earth is absorbed by gases such as carbon dioxide and methane in the atmosphere. This leads to a heating of the atmosphere as the radiation cannot escape back into space. The process is enhanced if more of these gases are present in the atmosphere.

(c) Outline possible strategies for reducing the amount of carbon dioxide that enters the atmosphere.

Carbon sequestration (i.e. capturing and burying the carbon dioxide deep underground); or increase the use of other energy sources such as solar, wind and nuclear energy.

Greenhouse effect



4. (a) Use the graphic on the right to explain how the natural greenhouse effect keeps our planet warm.

Radiation from the sun is partly absorbed and partly reflected back into space. Water vapour (in clouds or the air) as well as carbon dioxide and methane are molecules in the atmosphere that absorb radiation and warm our planet.

(b) Identify the cause of the enhanced greenhouse effect.

The enhanced greenhouse effect is due to human activities where additional carbon dioxide and methane are released into the atmosphere via the combustion of fossil fuels.

(c) Explain why the enhanced greenhouse effect is of major concern.

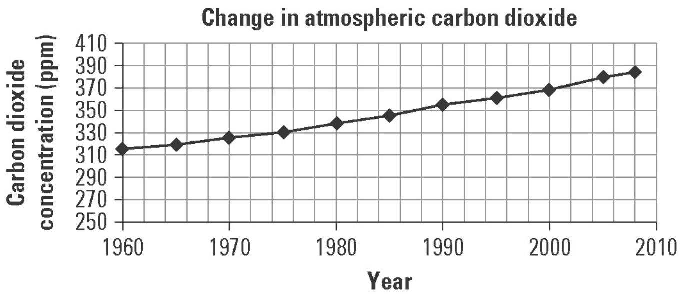
These additional amounts of greenhouse gas produce greater warming than normal.

Carbon dioxide concentrations

5. The following table provides atmospheric carbon dioxide data over recent times.

(a) Plot a line graph of this data and draw the line of best fit.

Year	CO_2 (ppm)
1960	317
1965	321
1970	327
1975	332
1980	340
1985	347
1990	356
1995	362
2000	370
2005	381
2008	386



(b) Global average temperatures have risen during this period. Is this evidence sufficient to state that global warming is caused by increases in carbon dioxide levels? Explain.

No the evidence is not sufficient on its own to make these conclusions. More data will need to be collected to confirm or refute the hypothesis.